

"Physics as Information Processing" ~ Chris Fields ~ Lecture 6

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Hello and welcome everyone. It is October 12th, 2023, and we are in the final lecture, session six in Physics as Information Processing with Chris Fields. So Chris, thank you for this journey. Exciting to get to this final lecture, which will be followed only by the last discussion for this go-around. So it's really awesome. And to you for this lecture. Thank you, Daniel. And welcome everyone to this last session that I'll be doing with a focus on biology. And just to give a brief review, as usual, we've focused in this course on generic quantum systems, by which I mean just any quantum system that has some number of degrees of freedom. And we really haven't made any assumptions about these systems other than the assumption that both interacting partners have sufficient degrees of freedom that they can be considered separable. So their boundary is small with respect to the total size of each of the interacting systems. And I want to emphasize again that the approach we've taken is completely topological. So we haven't assumed anything about embedding in space. And as we've characterized these systems, we focused really on two things. One, the amount of information that the two systems can exchange. And that amount is determined by the size of their boundary. It's determined by this number n , which is the Hilbert space dimension of the boundary. And as we've discussed now in several of the sessions, we can think of this boundary as a collection of n qubits via which the two systems A and B communicate. The other feature of the interaction on which we focused is the reference frames that are used by the two systems to interact with this set of qubits. And it's these reference frames that assign the information that flows between A and B meaning or functional significance or actionability or however you want to refer to the idea that when a system gets some information, it does something useful with it. So those are very generic kinds of ways of characterizing an interaction. You can think of it as a qualitative characterization, the semantics of the information, and a quantitative characterization, the amount of information. But other than that, we've left the nature of these systems open. And this characterization allowed us to reformulate the free energy principle in quantum information theoretic terms as the principle that interacting systems behave in a way that will asymptotically align their reference frames. And we saw that full alignment of reference frames is equivalent to entanglement. So the FEP drives systems toward entanglement, which is not surprising because that's what happens to generic quantum systems. They may start out separable, but if they interact long enough, they'll become entangled. So we've had a discussion that is really very generic. Now we showed a couple of sessions ago that under these very generic assumptions, we got some interesting outcomes.

And one is that if systems have sufficient degrees of freedom, under the action of the FEP, that they will be driven toward making a diverse set of measurements of their environment because they're trying to get information about it so that they can predict what it's going to do.

And that as they add ways of making measurements, they'll come up against a non-commutativity of their measurements.

And this may be due to intrinsic mathematical facts about their measurements, and it may just be due to their limited supply of free energy.

So either way, what this induces is compartmentalization and the requirement for internal classical communication on internal boundaries.

So I just wanted to reemphasize that this is a result of basic physics.

And we see compartmentalization, of course, ubiquitously in biology.

But this suggests to us that features of biology that are very familiar actually follow from fairly deep physics.

And compartmentalization is one of them.

We then talked last time about space and time and the computational requirements of on any system that is able to perceive an external clock or perceive an external spatial layout.

And we discussed this from a very kind of developmental psychology point of view.

We discussed it using essentially the kinds of terms on the right here, as opposed to the kind of language that's much more technical and mathematical that's used in physics to derive space time as emergent from information transactions.

But we at least hypothesize that both of these are ways of getting to the same sense of emergent space time.

And one important outcome of this discussion was that systems can perceive external clocks, so external periodic behavior with fewer computational resources, so before they're able to perceive spatial layouts.

And we'll get back to this later.

So organisms and maybe other complex systems seem to manage a collection of capabilities that are all mutually dependent.

So memory depends on having layouts of persistent things that can be distinguished.

So it depends on space, and it certainly depends on time.

Space gives structure to persistent objects.

Memory allows you to identify persistent objects.

And objects, in turn, give structure to space by being located in different places.

And objects enable the fundamental requirement for memory, which is some form of error correction, so that one has enough redundancy in the memory to check to see whether a memory has been degraded.

So organisms seem to manage a physically very important process, which is the use of space-time and memory as a basis for cognition.

You'll recognize these four functions that are mutually interdependent, as underlying complex cognition across the board.

And you can't really do complex cognition unless you have these capabilities.

So today we're going to put some of these things together in a discussion of biology,

and end by looking forward to future directions, not just in biology, but in all of quantum information theory. So let's talk about biological systems, living systems, which of course includes us, but also includes all other organisms and communities of organisms and so forth at multiple scales. And I want to put forward a fairly radical hypothesis that living systems are just generic quantum systems that the FEP has driven to a high level of computational complexity.

So what does this belief say?

It suggests that living systems are just quantum systems that have enough degrees of freedom and a sufficiently complicated internal dynamics that the FEP, by requiring them to predict what their environment is going to do next, has driven them to make complicated measurements and complicated inferences on the basis of those measurements that enable fairly complicated actions on the environment.

So these are systems that you would characterize as computers, essentially information processing systems, with a high degree of computational capability.

Now this is clearly as much a philosophical hypothesis as an empirical hypothesis, and that's due in part to the fact that we don't really have a good definition of what a living system is.

We can just point and say, you know, that's a living system and that thing over there isn't.

But beyond that, we don't have much of the theory.

And of course, people have worked on a theory of what is life for decades.

But there's not a lot of agreement.

And so this is, in a sense, a definitional kind of hypothesis.

And we test such kinds of hypothesis by seeing if they're useful conceptually in the theory to help us understand what living systems are doing.

Now, as soon as we put forward something like this, inevitably the question comes up, are living systems really quantum systems?

Or are they classical in some intrinsic sense?

And this question is really an artifact of two things.

One, the continuing association between quantum systems and microscopic systems, which is, of course, traditional.

Quantum theory was developed to deal with microscopic phenomena.

It was first tested as a theory of microscopic phenomena.

And it's only in the past couple of decades that quantum theory is being applied to big things, to cosmology, for example.

So there's a long tradition of association between quantum and microscopic and living systems, or mesoscopic, much bigger than atoms, bigger than molecules.

So, and they're not nearly as big as cosmological systems.

So they're somewhere in between.

And those systems have traditionally been regarded as classical.

And even chemistry, even biochemistry, is still taught in a very classical way.

So we're not used to thinking of these things as quantum systems.

But at least in the approach we've taken in this course, all physical systems are quantum systems.

And classical information exists only on boundaries, only on boundaries between systems.

And it only describes their interaction.

And I want to emphasize, I mean, I've said it before, but since this is the last class, I want to say it again, that this is not a universally held view.

It's a view that's becoming more common.

And I think a generation from now, it will probably be the dominant view of things.

But it's not fair to say that it's the dominant view now.

It's, I think, quite fair to say that it's a view on the rise, especially in the quantum information and quantum gravity communities.

But from a strict empirical perspective, we know a number of things.

We know that accurate molecular dynamics calculations require quantum theory.

We know that a lot of biologically important molecular processes use quantum coherence.

We know that things like hydrogen bonding can only be understood using quantum theory.

We know that cellular free energy budgets are way too small to support fully classical information processing.

If we think of information processing in terms of what proteins are doing or what other molecules are doing in terms of changing shape.

We know that shape changes in proteins and other molecules underlie all cellular information processing.

So if we think of them as actually implementing cellular information processing, we have way more information processing than could be funded classically.

And finally, we know that human cognition exhibits contextuality.

So human cognition uses quantum coherence.

It's not a classical causal process in all cases.

But this is still a very nascent field.

And it's a field with a lot of potential.

And I think, again, a generation from now, quantum biology is going to be regarded as a very mainstream endeavor, even though it's something of a small emerging field now.

And I'll just show you one picture in support of this third point.

This is a plot of measured free energy usage converted to numbers of bits processed at one kilohertz for a number of prokaryotic and eukaryotic cells,

all that I could find a few years ago that had good calorimetric measurements, versus the log of their area, their membrane surface area.

And the thing to note here is that these cells are processing classical information at pretty low rates, kilohertz to gigahertz rates for a thousand bits at a time.

And we know that a lot more is happening in cells than that.

Cells have many more than a thousand proteins.

And those proteins are all moving all the time.

And if that motion is processing information, then there's not nearly enough classical, not nearly enough thermodynamic power to power all of that information processing.

So let's now talk a little bit about life and living systems and what they look like.

Here's a view of living systems from the point of view of genome diversity.

This is a few years old, so it's out of date by now.

But it tells us a lot qualitatively.

It tells us that almost all genome diversity is microbial.

That the sorts of organisms that we're familiar with, medium-sized organisms like us, animals and plants, have only a tiny amount of genome diversity.

So we think of ourselves as pretty diverse.

We think of ourselves as quite different from other animals, from flies or worms or sponges or something like that.

But in fact, we're not all that different from them.

We share many, many proteins with all of those other organisms.

We share the same cellular organization and so forth.

We're not, in fact, all that different from plants.

And we're much more similar to the fungi than to the plants.

So we get a very biased view of genome diversity when we just look at familiar large types of living systems.

Most genome diversity is buried down there in the microbial world.

This isn't the only way we can look at life as a diverse set of organisms.

We can also look at it in terms of a cell lineage.

The fact that we're all related at the DNA level tells us that we're all descendants of a common ancestor, which is usually called Luca, the last universal common ancestor.

And Luca is generally modeled as being a cell, more or less like a bacterial cell, so fairly simple.

But the point of looking at life as a lineage is that we get a diagram that looks very much like the cell lineage from an embryological situation, much like the cell lineage of our own bodies growing from a fertilized egg.

And in particular, looking at life as a lineage shows us that not only do we share DNA with all other organisms,

we actually share a membrane, a cytoplasm, that's continuous in time all the way back to Luca, and so continuous in time between us and bacteria.

So if you look at life this way, you realize that life is in a sense all one organism, all one connected structure that's developed over time through division, cell division.

And so we're very closely related to all other parts of life in the same way that different cells in our bodies are closely related to each other.

The time span is just longer.

I mean, here are the time spans three and a half or four billion years between us and Luca.

These lineages have been diverging for that long, but they're still continuous.

And in particular, they form a continuous cell membrane.

So when you think of life in this way, we realize that we are reasoning from just one example when we think about terrestrial life.

And this strongly limits what we can say about living systems, what we can say about biology, that we really only have one example, the lineage that descends from Luca here on Earth.

And at some point, we may have more examples.

And there's certainly groups working on artificial organisms that aren't part of this lineage, that are constructed from parts derived from this lineage, but aren't members of this lineage themselves.

And that may tell us something new about life.

But if we think of life this way, we see something very important, which is that as life grows, its boundary gets bigger.

And since the boundary of a system constitutes its informational environment, its dimension getting bigger means that more information flows into it from its environment and more information flows out of it into its environment.

So as life develops from Luca, it gets a larger and larger informational environment.

And this dimension of its boundary grows and grows.

Now, what that means, if you go back to our original picture of two interacting systems, is that the number of the dimensionality of the physical environment is decreasing.

And we think of the physical environment as increasing in size.

We have the expanding universe and all that.

But what's increasing in size is space.

And if space is emergent, then that increase is not a real increase in degrees of freedom.

What we're really interested in is the Hilbert space of the physical environment as distinct from the Hilbert space of life.

And as life gets bigger, it literally, quite literally, eats the physical environment and turns it into more life.

So there's less physical environment around.

So the trajectory of life is toward a bigger informational environment and a smaller physical environment.

And as the physical environment gets smaller, you might imagine it gets easier to predict.

And as the informational environment gets bigger,

you have more information with which to predict the behavior of the physical environment.

So this all hangs together very nicely with the free energy principle.

Here's another picture of this same thing.

If we start with some boundary here on the left labeled B around some system labeled S, and it's embedded in an environment E,

then as S makes copies of itself through some process like cell division,

then it starts to fill up more of E.

It's actually converting degrees of freedom of E into degrees of freedom of its copies that it's made of itself.

And if we think of some larger boundary drawn somewhere in E,

here on the right side, I've labeled that B prime,

what happens is that B prime gets slowly filled up with copies of S,
and it starts to look like a large organism.
And that's what life is doing in the physical environment.
It's making a bigger boundary for itself at the expense of degrees of freedom of its physical environment.
So we can ask, what drives this process?
Why is life doing this?
Why is it making copies of itself?
And the answer to that question, at least one answer to that question,
is the free energy principle drives it to do this.
And I'll just show you some simulations from several years ago
of a cell population in this upper left panel.
The green cells are stem cells that divide,
and they've been embedded in an environment that's safe on the left side
and 100% lethal on the right side.
So they're embedded in an environment that's conducive to life on one side
and very dangerous as you get toward the middle.
And the purpose of these simulations was to study what happens
when the stem cells are able to divide and create not just other stem cells,
but create specialized progeny that don't divide.
So somatic progeny that don't divide and hence have free energy resources,
thermodynamic resources that they can devote to doing something other than dividing.
Cell division is incredibly expensive when it comes to energy.
So if you take that energy and apply it not to cell division,
but to protecting the stem cells,
you can allow the stem cell population to expand.
And that's what these other simulations show.
The blue cells are somatic cells that have very varying ability to protect the stem cells.
And if they're fairly good at protection,
and they're fairly hard to kill,
down in the right bottom quadrant,
you can actually get a population expanding well into the highly lethal part of the environment.
So this shows you the FEP in action.
What the stem cells are doing
is constructing a very predictable neighborhood
around themselves in the environment
by dividing to make progeny that don't themselves divide,
that just sit there
and help protect the stem cells.

So the stem cells are now living in a safer, more predictable environment, and that allows them to invade more and more of their physical environment by having this fairly benign informational environment.

So this is the FEP in action, driving the expansion of life at the expense of the physical environment.

So that lets us go back to this question that was raised last time about the mechanism of these scale transitions that we see ubiquitously in life.

Life gets bigger and bigger as time goes on phylogenetically and as time goes on developmentally.

We start with some molecular pathways that somehow get assembled into a cell.

That's the origin of life question.

How does that happen?

And about 2 billion years later, cells start to group together to make multicellular organisms.

Now, bacteria group together to make facultative multicellular systems like microbial mats 3 billion years ago, very, very early on.

But we don't see the emergence of obligate multicellulars until much, much later, before the Cambrian explosion, but still less than a billion years ago.

And from there, things get more complicated very quickly.

So one has large multicellular organisms.

They develop complicated social relations that probably first emerged with social insects, but are certainly well-developed in invertebrates, including in mammals.

Organisms group together in very complex ecosystems.

And recall here that it's now known
that all multicellular organisms
have obligate microbial symbionts.
So all of our bodies are, in fact,
very complex ecosystems
populated by many, many different species of bacteria.
And that's true not just for us,
but for trees and insects and worms
and sponges and everything else.
So organisms,
from a body point of view,
are themselves ecosystems.
And organisms organize themselves into ecosystems.
And these ecosystems come into contact
across the entire biosphere.
So processes as simple as the wind
disperse organisms as small as bacteria
all over the planet.
So this scaling of life
toward larger and larger organizational structures
is, in a sense,
the major thing for biology to explain.
And the hypothesis that we floated last time
was that this transition
from very small scales
to very large scales
uses the same mechanism over and over again,
or a mechanism that has the same structure
over and over again.
So the renormalization group
is trivial, basically.
It's just the identity
operating at different scales,
or mapping from one scale to the next.
So the question is,
can we approach this in any way?
Does this make sense?
And to try to make sense of it,

we can go to the one real developed theory
that we have in biology,
which has been formulated
by Dawkins and others
as universal Darwinism.

This is just kind of an abstraction
of the modern synthesis
between Darwinian evolution and genetics.

But universal Darwinism
is incredibly simple.

It's very obvious.

It says,

if you have systems
and they can copy each other,
they can copy themselves
and the copies can diversify,
then if you want to know
what's going to happen
at the population level,
what you will find
is the systems
that copy themselves
and diversify most efficiently
will win

because there'll be more of them.

So they end up dominating the population.

And winning gets called natural selection.

But natural selection

is not an extra part
of the Darwinian process.

It's just the outcome
of efficiency,

of the systems
that copy themselves
most efficiently

coming to dominate
any population.

So there's no sort of

kind of hand of God
or whatever reaching down
and selecting some organisms
and not others.

And of course,
when Darwin put
the idea of natural selection
together,
he was thinking
very explicitly
of the analogy
of a gardener
or a farmer
selecting the crops
or the animals
or whatever
that the gardener liked best
and keeping those.

And that's where
this selection metaphor
comes from.

But there's no real process
of selection
in evolution
by natural selection.

It's just copying
and diversifying.

But this clearly
is just the outline
of what a real
mechanistic theory
would be
because we need
to have a story
about what copying
is
and what diversifying
is

and what efficiently
means.

And in particular,
to address the question
that we raised
a couple of slides
ago,
we need to know
whether these mechanisms
are the same
or at least
isomorphic
of the same form
at different scales.

So,
that's what I want
to talk about
for the next little bit.

let's start
with copying.

Up until the 70s,
copying
in biology
really meant
DNA replication
and
to some extent
cell division.

replication,
but the focus
in evolutionary
theory
was on

DNA replication.

And in the 70s,
it was realized
that
whole genes

could be
duplicated,
so whole genes
could be copied.
And then
shortly
after that,
it was realized
that particularly
in prokaryotes
where genes,
functionally related genes
are next to each other
and chromosomes
often
in what are called
operons,
that
operons
could be duplicated
and so entire pathways
could be duplicated.
And as genomes
began to be sequenced
and structures
of genomes
were understood,
it became clear
that genomes
could be copied.
And
of course,
we have two copies
in each of our cells
with the exception
of red blood cells
of our complete genome.
But

some plants,
for example,
have six or eight
or even more copies
of their genome
in each of their cells.

So
there's
large-scale
genome duplication.

Cell division

is
clonal duplication
of cells.

Organismal reproduction
works by cell division
in all known systems.

Colonies or societies
of organisms
reproduce themselves
in part
through cultural transmission.
and entire
niches reproduce
themselves.

For example,
in processes
like
forest evolution
or desertification
or even
urbanization,
we see
humans reproducing
the same sorts
of niches
in city
after city,

for example.

people.

So

these forms

of copying

have the kind

of hierarchical

dependencies

that one might

expect.

So to duplicate

pathways,

you have to

duplicate genes.

To duplicate

cells,

you have to

duplicate the genome.

To duplicate

organisms,

you have to

duplicate cells.

and this

suggests

that this

copying operation

is indeed

scale-free.

That

each

level,

the copying

operation at each

level enables

the copying

operation at the

next level.

Now,

with diversification,
we see much
the same thing.
In the early
days of
the modern
synthesis
and still
in the
work of
people like
Richard Dawkins,
diversification
is generally
thought of as
random.

So mutation,
for example,
is thought of
as random.

And
here I have
Barbara McClintock,
who is kind
of the hero
of non-randomness
in
doggedly
over decades
pointing out
that
much of
genome
diversification
is not
random at
all.

And

that
genomes
actively
diversify by
moving whole
clusters of
genes around
and
duplicating
them or
just changing
their positions
so that
they're under
the influence
of different
local
control.

gene and
genome
level.

You have
active
diversification
by various
processes.

Enzymatic
processes

or, as

McClintock
showed,

is transposon
mediated
mechanisms,

which are
enzymatic

at the
bottom.

Cells
actively
diversify by
lots of
different
mechanisms.
systems.

And in
the 70s,
Carl
Woes
was, in
part,
personally
responsible for
pointing out
that in the
microbial world,
there's a lot of
horizontal gene
transfer.

and, again,
no one believed
him for quite
a while.

But now it's
broadly accepted
that microbes
exchange DNA
very frequently
and across
large distances
phylogenetically.

And this is
how things
like microbial
resistance to
antibiotics

spread so
quickly,
or one way
that it
spreads so
quickly.
And it's
a tremendous
agent of
essentially
speciation
to the extent
that that
even makes
any sense
in the
microbial
world.
Of course,
in our
cells,
we have
functional
differentiation.
As mentioned
earlier,
there are lots
of symbiotic
interactions
that lead
to diversification
in cells
by radically
altering their
local
environments.
And, of
course,

sex is a
mechanism for
diversification.

Organisms
actively
diversify in
many different
ways.

The most
obvious one
is just by
moving around,
by moving to
a different
environment.

The organism
is exposed to
different inputs
and is allowed
different outputs.

And so it
can move into
a different
direction,
morphologically,
genetically,
etc.

organisms can
regulate their
own
evolvability
by altering
things like
DNA repair
systems or
activating
transposons.

Again,

these are
sorts of
things that
went very
much against
the grain when
they were first
discovered but
are now pretty
well accepted.
And, of course,
organisms can
learn and so
can diversify
their behavior
by diversifying
their computational
capabilities.

And social
groups, of
course, actively
diversify in
many different
ways.

And organisms
are increasingly,
non-human
organisms are
increasingly regarded
as having cultures
that they pass
on intergenerationally
by intergenerational
learning and can
exchange with
other groups.

So,
diversification

is active
and occurs
at every
scale.

And,
again,
diversity,
although this
is a bit
less clear
than in the
case of
copying,
diversification
mechanisms
at one
scale enable
diversification
mechanisms
at other
scales.

So, for
example,
transposon
mediated
mechanisms
enable
horizontal
gene
transfer
in
microorganisms.

And the
FEP
tells us
that efficiency
is actually
about

intelligence.
because the
FEP
drives
systems to
build good
models of
their
environments
or the
behavior of
their
environments.

And what a
good model
lets a
system do
is reach
a single
goal by many
different means
because it
allows, in a
sense,
planning.

and it
allows
flexibility
and adaptability.

And I put
William James
here on this
slide because
he was the
first, as far
as I know,
to define
intelligence as

this ability
to use
different means
to reach the
same goal.

Now, in
evolution,
that's long
been known,
and it's
called
convergent
evolution.

When you
see two
lineages
adapting to
a single
environment
with very
different
genetics and
very different
morphologies,
and often
the organisms
will end up
with similar
morphologies
that work
well in that
environment,
but very
different
genetics and
very different
biochemistry.
So,

throughout
evolution,
we see this
phenomenon of
functional
convergence
balancing
structural
divergence at
many, many
different levels.

So, we
have
multiple
genes for
very similar
proteins.

We have
multiple
pathways that
do more or
less the
same thing.

We have
different cell
types that
support the
same basic
functions.

We have
tissues that
have, that
are fairly
different, but
that have the
same functions.

For example,
eyes and

different kinds
of organisms
that allow
vision but have
very different
structure.

And, of
course, we
have functional
convergence and
social relations.

And if we
look at how
cells interact,
traditional
evolutionary
theory really
focuses on
competition.
evolution.

But we
now know that
cells cooperate,
that they
exchange resources
in various
ways, even
across vast
differences in
phylogeny.

For example, the
interchange between
plants and
fungi, or
between bacteria
and us.
cells, we
know that

cells, in
a sense,
persuade each
other by
secreting
things like
hormones that
cause other
cells to do
different things.

Cells
deceive each
other.

For example,
microbes that
are trying to
escape the
immune system.

And cells
coerce each
other.

Cells even
kill each
other when
they detect,
for example,
genetic
defects.

So, these
sorts of
social and
economic and
even political
ideas are
useful for
describing living
systems from
the level of

cells on up.

They were
developed in
social settings
to describe
what goes on
in human
societies and
then extended
to other
animals.

And now
it's very
clear that
they extend
to individual
cells.

So, when
we think about
social and
intelligence,
we can think
about social
intelligence even
at the level
of individual
cells.

And we
see these
sorts of
competition,
cooperation,
resource exchange,
etc., even at
the level of
microbial mass,
so biological
structures that

are 3 billion
years old.

Now, of course,
intelligence is
necessary, but
it's not sufficient
for being
selected.

protected.

It won't
protect you
from impacts
of asteroids
that happen to
land on your
planet, which
has happened
often in the
evolution of
the Earth.

And we're
now trying to
figure out how
to predict and
prevent such
things.

But it's, you
know, who
knows whether
we'll be
successful at
that.

intelligence is
necessary when
it's viewed in
this FEP sort
of way.

Intelligence is

what lets you
build models
that enable
you to predict
the environment.

So where does
this leave us?

By thinking
of evolutionary
theory in
terms of
scale-free
mechanisms, we
seem to have
a picture in
which we start
with quantum
systems with a
certain level of
complexity, and
the FEP actually
gives us scale-free
biology, or
something very
close to scale-free
biology.

And that's very
interesting because
the FEP, of
course, is a
generic principle.

It just says that
systems do what
they need to do to
keep existing.

And this way of
thinking gives us
some predictions.

One is that we'll find compartmentalized problem-solving everywhere because the FEP drives compartmentalization, and the reason is just non-communativity of measurement processes.

it predicts that we'll see hierarchical organization everywhere in the same way we did on that first compartmentalization slide where once you have a number of compartments, you need a meta compartment that, if nothing else, distributes free energy.

So some sort of very basic attention system.

And we see attention systems, of course, throughout biology.

technology.

It predicts that living systems will have very limited classical data structures.

So since they have

to write classical data on their boundaries, and their boundaries are much smaller than they are, then they'll, and because writing classical data is expensive, that living systems will tend to use as little classical data as they can get away with.

Since classical data is needed for thermodynamic irreversibility, where we see thermodynamic irreversibility, that's where we look for classical data.

We saw that organisms will be able to detect time first before they can detect space.

So we can expect organisms to have clocks even before they can represent space.

And of course, we see things like diurnal variation throughout biology.

And finally, the FEP predicts this

asymptotic collapse
of separability back
to entanglement.
So it predicts that
living systems will
eventually degrade and
become indistinguishable
from their
environments.

And we know that
happens at the cell
level and the
individual organism
level.

You know, we call
it death.
death.

So the FEP seems to
give us a lot of the
features of biology.
And it gives us these
features in a scale-free
way.

And it derives them
from this very basic
physics.

So where does that
leave us?

I think it leaves us
here with a challenge,
not just to thinking
about the FEP,
which is what this
course has been
about, but to
quantum information
theory in general.
And that challenge is

to evolve from the kind of conceptual and theoretical utility that we've been discussing in this course, and which is very well developed in the physics community, to some sorts of empirical and technological utility. technology.

And I've listed three areas here where QIT is very active, one in quantum gravity and cosmology.

And you're probably familiar with things like string theory that sort of fairly famously seem not to be empirically testable. people.

And the quantum gravity community is enormous, many thousands of people working in this area.

But coming up with theories that can be tested empirically at the scales that we can actually do experiments, so at accessible energy scales, for example, has proved to be very difficult.

So this is a real challenge to the field, is somehow getting not just to empirical testability, but to some sort of technological utility. And you can imagine that if the main prediction of quantum gravity is that space-time is emergent, that there could be enormous utility in that.

Some ability to manipulate the way space-time emerges. The second area, of course, is quantum information in computing.

And here, massive resources are being put into technological utility.

And I think it's reasonable to expect that we'll have useful quantum computers within a generation, at any rate.

So there, I think, the field is going quite well.

And the last area,

what we've talked
about here, is
quantum biochemistry
and biology.
technology.

And this, I think,
is an area with
enormous promise
that maybe some of
you will actually
contribute to
making this area
more useful
technologically and
more accessible
empirically.

at least we have a
picture of how
these different
areas fit
together.

So that's
it.

Thank you very
much for
participating in
this class.

Andrew will be
doing...
well...
Sorry.

Sorry.
Could you just
repeat the last
30 seconds from
and or will be...
Chris?

Andrew will be
at his last
discussion session
on Saturday,
the 20th of
October, at the
usual time, 8 a.m.
Pacific time.

And the interactive
Q&A is open.

I really appreciate
your questions.

I try to answer
them.

And thank you
very much.

Awesome.

Can I ask a
question or do
you want to call
it here?

Sure.

Go ahead.

All right.

Inks wrote,
Chris spoke last
time introducing
the mechanisms of
scale transition.

I'll read your
question and then
I'll fix the
cropping on the
live stream.

Chris spoke last
time introducing
the mechanisms of
scale transition

of these
essentially semantic
theories about how
one scale embeds
in another scale.

I would love to
learn more about
this, especially
in terms of
natural language
semantics and
human cognition
semantics versus
machine learning.

Wow.

Well, that's a very
good question.

Oh.

I think that's a
very relevant
question given
the sorts of
machine learning
that we see
deployed in
systems like
chat GPT.

And especially
as these
systems start
to associate
language with
images, for
example, and
as such
systems are
embedded in
robotic

systems that
can actually
move around
in the world
and interact
with real
objects.

I think
one of the
key questions
in natural
language semantics
is to what
extent does it
depend on
reference?

To what
extent do
the meanings
we attach
to words
or the
usages
that we
allow
for words
depend on
our ability
to connect
words to
external
objects?

And to
what extent
do those
abilities
rely on
connections

between words
themselves?

So if you
think of an
LLM like
chat GPT,
by itself,
it's just a
statistical
model of
relationships
between words
as used
in some
corpus of
text, which
is large,
of course,
but we
can debate
its
representativeness.

And so
that's a
model of
semantics
that's entirely
self-contained.

It's entirely
relations between
different pieces
of language.

Whereas when
we start to
connect those
words,
statistics,
to images

or better
yet to
actual
objects
that are
interacted
with not
just visually
but haptically,
for example,
by a robot,
they start to be
connected to
external reference.

And
that
transition I
think is going
to be very,
very interesting
to watch.

and it's
just starting.

So we all
have ringside
seats on
an incredible
experiment in
semantics.

And
so
that would
be my
response really
to that
question is
keep your
eyes on this

experiment
and see how
it goes
because it's
bound to
tell us
something about
how
semantics
really works.

And I
suspect that
there will be
enormous
opportunity to
start relating
the results of
that experiment
to what we
see in
developmental
psychology,
for example.

Wow.

Awesome.

All right.

Tucker asks,

Areas of
macroscopic
quantum
mediated
phenomena in
biology that
get Chris
Fields excited?

I think
the one that
excites me the

most right
now is
just this
question of
how cells
are managing
to do the
information
processing that
they're doing
and
how much
of it
they can
accomplish
or
I think
the right
way to put
it is
how can
they get
away with
so little
classical
information?
And
I don't
think we
have the
cell biology
yet to
answer that
question,
but I
think asking
the question
may help

us design
the right
kinds of
cell biology
to look
for an
answer.
And our
cell biology
has been
organized around
the idea
that the
structures that
we see
are essentially
static,
that they're
classical,
and that
they're
constructed in
a way that's
thermodynamically
irreversible.
But if we
think of
what we
measure as
not
necessarily
reflecting a
static
structure,
I think
we'll start
to
conceptualize

it differently
and build
models of it
differently.

I mean,
clearly when we
look at things
like protein
structure,
we're looking
at snapshots
of something
that's very
dynamic.

And when
we look in
a chemistry
textbook and
we see
pictures of
this
conformation
that changes
to that
conformation,
we think of
those as
static,
stable
structures,
that that's
kind of a
switch.

But it's
not actually
a switch the
way a light
switch is.

It's a much
more dynamic
process.

And the
kinds of
molecular
dynamic
simulations
that allow
you to
study that
process are
incredibly
complicated
and time
expensive
and uncertain
in resolution
problems and
other things.

But I
think thinking
of these
as kind
of fluid
reversible
processes
is going
to change
the way
we think
about them.

That's a
great point
and it
reminds me
of the
more

physics
based way
of thinking
about the
paths that
could happen
between
observations
or the
statistics
based way
of samples
of a
dynamical
process.

And in
biology,
we know
that even
if we take
a snapshot,
okay,
this organism
is one
centimeter
today,
it's two
centimeters
tomorrow,
something
changed.

So we
know there
is a
developmental
process,
then we
know there's

an ecological
process,
and then
there's an
evolutionary
context,
eco,
evo,
devo.

And then
from that
more biological
understanding
of contextual
developing
organisms,
all of
these different
biological
phenomena
actually help
us expand
our space
of measurement
and analysis
for material
that on
one hand
some people
may have
called
inanimate,
but then
they have
their own
dynamics,
different
time scales,

and so
bringing some
of those,
like the
challenges
of measuring
and observing,
intervening in
historical
and biological
things,
gives us a
bigger
understanding
even on
measuring
and summarizing
and analyzing
ensembles
or scales
that don't
immediately
appear to
have that
kind of
historiosity
like pieces
of silicon
or carbon.
Yeah,
that's well
put,
and I
think we
have the
same problem
as we go
up in

scale also.

We tend

not to think

of things

like societies

as living

systems,

but in

one sense

they clearly

are,

and to

the extent

that we

can think

of these

structures

that are

much larger

than we

are

individually

as

living

systems,

cognitive

systems,

et cetera,

I think

we'll

understand

their

dynamics

better.

Okay,

final question.

Upcycle Club

wrote,

how can
a quantum
system
detect
and respond
to quantum
context
shifts,
switches,
which are
changes in
the measurement
basis or
the entanglement
structure of
the system?

Well,
the answer
here is
simple and
not very
informative.

It's by
making
measurements,
which is
all that
quantum
systems can
actually do.

And
making
measurements
that follow
probing
actions
that
try to

push the
boundaries
of the
model of
a context.

context.

You know,
when having
this discussion,

I always
like to
go back
to the
frame
problem

in
artificial
intelligence,
which is

the
problem
formulated

in the
late
60s

of
predicting
what won't
change as
the result
of an
action.

so it's
sort of
the inverse
of the
problem
of predicting

the side

effects.

And if you

can predict

all of the

side effects,

you can

predict what

doesn't

change.

So it's

equivalent to

the question

of predicting

all the

side effects.

And context

change is a

side effect.

And it's

often an

unobserved

and maybe

unobservable

side effect.

that if

you have

a model

that says

I'm working

in this

context,

then you

can

intentionally

probe the

boundaries of

that model

and see
if you
can actually
alter the
context you're
in as a
way of
discovering
what context
you're in.

And,
you know,
humans do
that in
social settings
all the
time.

But I
think this
is something
that we
need to
think about
happening at
many different
scales.

I suspect
that cells
are doing
this all
the time.

We just
don't know
how.

There's so
much to
say there.

It just

makes me
think of
the paradox
or the
tension
between
expecting
the expected,
expecting
the unexpected,
and life
needing to
expect both
of that.

Right.

So,
thank you,
Chris.

It's been an
amazing lecture
series.

Many people
in the live
chat are
very appreciative
and we'll
look forward
to their
submitted
questions and
their
participation
in the
final discussion
coming soon.

So,
thank you,
Chris.

Till next time.

Okay,

and thank you

all.

Bye.

Ciao.

Bye.

Thank you.